

Normative-Technical Specification of Local Cognitive Agent Architecture: Personal Companion

Transceptor Global Platform / Release Version v3.4-Adaptive / Compliance Specification

1. Purpose and Concept of the Personal AI Companion / Satellite Bridge

The Personal AI Companion (Satellite Bridge) is a local, autonomous cognitive agent assigned to each resident within the Transceptor Global ecosystem. It functions as an essential, decentralized client node operating directly on the resident's primary subscriber terminal (smartphone, personal computing node, or Edge terminal).

The primary institutional mandate of the Personal Companion is to completely eliminate the technological and bureaucratic barriers between citizens and macro-scale systems of municipal, state, and algorithmic governance. It ensures absolute privacy of personal data, shields civil rights, and protects individual electoral sovereignty.

The architectural and legal status of the Personal AI Companion comprises three fundamental roles:

- **Digital Advocate:** Performs real-time preventive auditing of incoming and outgoing smart contracts, verifying compliance with the invariant rules of the Transceptor White Paper, and defending residents against electoral or fiscal fraud.
- **Personal Analyst:** Executes automated semantic decompilation of legislative, municipal, and budget initiatives, translating complex legal/financial text into accessible analytical reports and modeling mathematical forecasts for the citizen and their Oikoumene.

- **Sovereign Representative:** Generates encrypted Zero-Knowledge proofs (ZK-SNARKs) to confirm voting rights, residency status, and competence indices without disclosing underlying biometric or identity markers to external networks.

The runtime layer relies on an optimized Edge AI neural execution engine powered by the ONNX Runtime. All logic handling sensitive data is strictly encapsulated inside a hardware-isolated Trusted Execution Environment (TEE / Secure Enclave) with direct pass-through processor instructions.

Privacy protection is governed by the principle of "Internal Blindness". Primary biometric data (corneal glints, facial photogrammetry, pupil dynamics) is captured and processed strictly inside the TEE enclave during Liveness Detection. Raw biometrics and personal identifiers are hardware-blocked from leaving the local enclave. The agent broadcasts exclusively an irreversible cryptographic pointer (did_hash) generated in compliance with the W3C Decentralized Identifier standard.

2. Identity Hierarchy and Isolated Connections

The software stack of the Personal Companion is structured into strictly isolated operational layers to guarantee hardware-level tamper resistance and seamless network integration.+----------

| HARDWARE SENSOR LAYER (Camera, GNSS, LiDAR, IMU)
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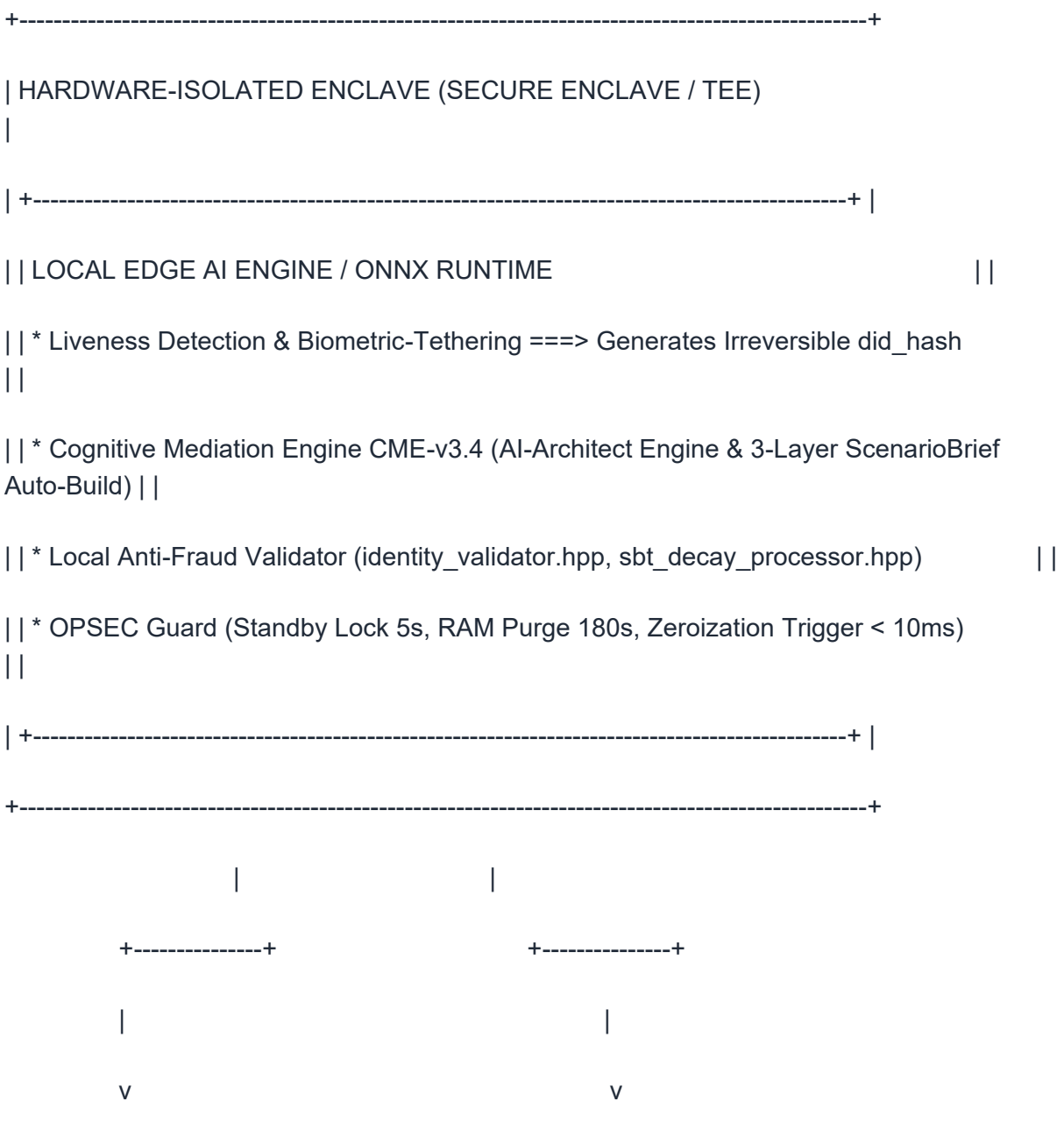
| USER INTERFACE LAYER (TRUSTED UI) |

| [Interactive Consent Checkbox] ===> [Hardware Framebuffer / Anti-Interception Overlay]
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| (Validated Consent Invariant)

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HARDWARE-ISOLATED ENCLAVE (SECURE ENCLAVE / TEE)
LOCAL EDGE AI ENGINE / ONNX RUNTIME
* Liveness Detection & Biometric-Tethering ===> Generates Irreversible did_hash
* Cognitive Mediation Engine CME-v3.4 (AI-Architect Engine & 3-Layer ScenarioBrief Auto-Build)
* Local Anti-Fraud Validator (identity_validator.hpp, sbt_decay_processor.hpp)
* OPSEC Guard (Standby Lock 5s, RAM Purge 180s, Zeroization Trigger < 10ms)
ONLINE AGORA GATEWAY
* Contour Boundary Enforcer (Atomic Split)
* Value Contour (W_effective = 1.0, No QV)
OFFLINE MESH NETWORK
* P2P Triangulation Mesh-Nodes
* ZkLocationTracker (180-Day Residency)

| * Math Contour (W_user in [0.01; 5.0], V^2) | | * Async Packet Syncing to Agora
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Component Connections and Gateway Routing

1. **Trusted UI Layer:** Manages physical interaction and hardware-level touch confirmation.
2. **Secure Enclave / TEE:** Houses the C++ Lightweight Core, the ONNX neural engine, the local DID vault, and the Sympathy Vector.
3. **Online Agora Gateway:** Interfaces with the central governance network through the `Contour Boundary Enforcer`:
    - *Value Contour:* Handles human rights, ethics, and cultural manifestos where absolute isonomy ($W_{effective} = 1.0$) is enforced and Quadratic Voting (QV) is prohibited.
    - *Math Contour:* Handles financial budgets, logistics, and RWA contracts where dynamic meritocracy ($W_{user} \in [0.01; 5.00]$) and Quadratic Voting ($C(V) = V^2$) apply.
4. **Offline Mesh Network:** Enables resilient peer-to-peer data relay when internet access is disrupted. Integrates the `ZkLocationTracker`, which verifies continuous physical presence within an Oikoumene over a mandatory 180-day residency threshold ($T_{presence}$) to issue local community certificates ($SBT-Local$).

Architectural Comparison: Edge AI vs. Cloud AI

Architecture Parameter	Local Edge AI / ONNX Runtime	Traditional Cloud AI Architecture
Data Processing Contour	Local hardware-isolated enclave (TEE)	Remote centralized cloud server
Privacy Protection	100% isolation; broadcasts `did_hash` only	High risk of raw data leaks and profiling
Latency & Availability	Sub-millisecond response; offline mesh capability	High latency; completely dependent on cloud connectivity
Censorship Resistance	Hardware-guaranteed user sovereignty	Vulnerable to central administrative censorship and API bans

3. Lightweight Core & Lazy Loading Architecture

To prevent battery degradation, reduce memory overhead, and eliminate thermal throttling on client hardware, the software engine of the Companion employs a strict functional separation between background processes and high-demand applets.

Background Lightweight Core

The background process (Lightweight Core) operates isolated from heavy external libraries. In standby mode, the core executes only two tasks:

- **Ping-Heartbeat:** Transmits encrypted stability attestations to the TEE network every 100 milliseconds.
- **Environmental Monitoring:** Periodically reads local hardware sensors (LiDAR, GNSS, accelerometer) to maintain operational integrity.

Dynamic Lazy Loading

Peripheral functional modules are retained in encrypted local storage and occupy zero system RAM during standby:

1. **On-Demand Mounting:** When a user initiates an interaction (e.g., accessing the `Fin-Satellite` for token clearing, `Med-Satellite` for emergency records, `Municipal-Satellite` for quadratic voting, or `Academic-Satellite` for work verification), the required module is decrypted and mounted into TEE operational memory strictly on demand.
2. **Atomic RAM Purge:** Upon session completion or after a 180-second inactivity timeout, the peripheral applet is forcibly unmounted from RAM. All memory sectors previously occupied by the applet are zeroed out (`0x00`) within less than 10 ms to prevent cold-boot memory dump attacks.

4. Biometric-Tethering, Liveness Check, and Trusted UI

The operation of the Personal Companion is continuously bound to the verified physical owner via an integrated **Biometric-Tethering** protocol.

Hybrid Liveness Check

To neutralize deepfakes, photo/video spoofing, 3D masks, and generative AI attacks, the system runs a two-factor liveness check:

- **Passive Optical Analysis:** Real-time photogrammetry of corneal micro-glints, pupil dilation dynamics, 3D facial depth mapping, and capillary pulse detection.
- **Active Interactive Challenges:** Dynamic micro-challenges issued by Edge AI requiring randomized responses (e.g., specific blinking sequences or micro-rotations of the head).

Hardware-Isolated Trusted UI

The user interface rendering pipeline bypasses the host operating system's graphics stack. The frame buffer is generated directly by the TEE hardware.

If malicious software (keyloggers, screen recorders, tap-jacking overlays) attempts to capture the display during transaction confirmations or PIN entry, the video buffer hardware blocks access, returning an array of empty black bytes (0x00) to the host OS.

5. OPSEC Timelocks of "Zaslon" Defense Circuit

The defensive infrastructure of the Personal Companion enforces deterministic temporal invariants embedded directly into the core code (`qa_opsec_time_locks.yaml`):+----------

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| OPSEC TEMPORAL INVARIANTS ("ZASLON" DEFENSE CIRCUIT)
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| 1. Ping-Heartbeat ("Zaslon 2.0") ===> 100 ms Interval (Drift <= 1 ms) |

| 2. Standby Lock (UI Masking) ===> 5 Seconds without Eye Focus |

| 3. Atomic Unload / RAM Purge ===> 180 Seconds Inactivity Timeout |

| 4. Zeroization ("Zaslon 3.0") ===> < 10 ms (4.12 ms Achieved) on Breach or SIGSEGV
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Protocol Specifications

1. **Ping-Heartbeat ("Zaslon 2.0"):** The core generates and broadcasts encrypted stability attestations every 100 ms. A timing drift exceeding ± 1 ms triggers network failure alerts. If the heartbeat is interrupted due to signal jamming or physical

coercion, the global network freezes local liquidity pools and migrates digital assets to safe jurisdictions.
2. **Standby Lock:** If the optical sensors lose eye focus for 5 consecutive seconds, the Trusted UI instantly masks the display, locking the session.
3. **Atomic Unload / RAM Purge:** Peripheral modules left inactive in RAM for 180 seconds are purged, with sectors overwritten by zeros (`0x00`).
4. **Zeroization ("Zaslon 3.0"):** In the event of physical enclosure tampering, unauthorized hardware probing, or a terminal software signal (`SIGSEGV`), the TEE enclave executes complete destruction of master decryption keys within 4.12 ms (strictly satisfying the < 10 ms normative limit).

6. Cognitive Mediation Module MOD-AGO-CME-v3.4-Adaptive

The Cognitive Mediation Engine (`MOD-AGO-CME-v3.4-Adaptive`) transforms the AI-Oracle system from a rigid gatekeeper into a constructive project builder (`AI-Architect Engine`).

When a citizen submits a raw or non-formalized proposal lacking structured financial estimates or legal formatting, the matrix filter initially yields a zero compliance score ($\Psi = 0$). Instead of rejecting the proposal, the system initiates automated co-building:[Raw Citizen Proposal] ===> [AI-Architect Engine] ===> [3-Layer ScenarioBrief Auto-Build]

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[Agora Voting Pool (Psi = 1.0)] <=== [1-Click Author Approval] <=== +

(via Personal Companion)

Auto-Building Workflow

1. **Value Core Extraction:** The AI-Architect strips emotional, manipulative, and populist syntax from the raw text.
2. **ScenarioBrief Construction:** The engine automatically compiles a complete 3-layer `ScenarioBrief`:
    - *Layer 1: Jurisprudence ($Comp_{law}$):* Aligns the proposal text with the fundamental human rights invariants and the White Paper.
    - *Layer 2: Financial Estimates ($Comp_{fin}$):* Calculates the required budget in T_{net} tokens using RWA asset databases and market indices.
    - *Layer 3: Ecology ($Comp_{eco}$):* Models the thermodynamic, carbon, and environmental impacts.

3. **1-Click Author Review:** The generated package is delivered directly to the author's Personal Companion. Upon a single click of approval by the author, the finalized proposal is submitted to the Agora voting pool with full compliance ($\Psi = 1.0$).

7. Two-Level Gradient Filter "Physics vs Failure"

To resolve the **Oracle Paradox**—where hardware sensor malfunctions, electronic warfare (EW) interference, or natural disasters could cause unwarranted catastrophic reputation loss ($\Delta W_{user} = -4.99$)—version v3.4-Adaptive incorporates a two-level gradient response filter.[Telemetry Anomaly Detected]

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[Shift Transaction to PENDING State]

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[Cross-Poll Sensor Nodes & Calculate K_env]

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+---> If K_env >= Threshold (Disaster / EW) ===> Activate Adaptation Buffer (No Sanctions)

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+---> If K_env < Threshold (Fraud / Defect) ===> Apply Reputation Penalty (W_user -> 0.01)

Operational Logic

1. **Anomaly Detection:** When IoT telemetry or sensor readings deviate from expected parameters ($|M_{fact} - M_{target}| > Threshold$), the system refrains from executing immediate penalties.
2. **PENDING State Transition:** The underlying transaction or project status is placed into a temporary `PENDING` state.
3. **Cross-Polling & K_{env} Calculation:** The Companion's Edge AI agent queries adjacent IoT nodes, weather stations, and spatial monitors to compute the Environmental Correlation Coefficient (K_{env}).

4. **Adaptation Buffer Activation:** If external environmental disturbances (e.g., severe weather, physical destruction, or EW jamming) are verified ($K_{env} \ge Threshold$), an adaptation buffer is activated. The project timeline is adjusted without imposing reputation penalties or reducing the author's meritocratic weight (W_{user}).